

Detim Zhao

✉ detimzhao@gmail.com

🌐 [linkedin.com/detimzhao](https://www.linkedin.com/detimzhao)

🐙 github.com/DetimZhao

Education

Bachelor of Science, Software Engineering

May 2026

Arizona State University, Polytechnic – GPA: 3.88 / 4.00

Mesa, AZ

- Barrett (Honors), Dean's List (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2025)

Technical Skills

Languages – Java, Python, C, C++, PostgreSQL, MySQL, Bash/Shell Scripting, UML

Technologies – pandas, scikit-learn, matplotlib, seaborn, NumPy, PyTorch, Transformers (Hugging Face), OpenCV

Tools – Docker, AWS, Git, GitHub, Unreal Engine 5 (UE5), LaTeX, Jupyter Notebook, PlantUML, Astah, Photoshop, Illustrator, Excel

Methodologies – Scrum, Agile Development, Git Workflows, Branching Strategies

Experience

Extended Reality (XR) Creative Developer

July 2024 – Present

Meteor Studio

Mesa, AZ

- Containerized a GPU-enabled multi-dependency environment using Docker and Docker Compose, resolving complex and inconsistent Linux/WSL2 build and reducing setup time from an hour to 16 minutes.
- Devised a proprietary volumetric rendering and camera calibration pipeline, leveraging a two-stage 3D point cloud registration process (RANSAC + ICP) for precise extrinsic alignment. (Python)
- Developed a research prototype in Unreal Engine 5 (UE5) to simulate a stereoscopic-camera-equipped drone autonomously following a runner on a track, utilizing a custom UI for fine-tuning motion control.
- Co-authored two research posters on volumetric rendering and camera calibration, presented at ACM HotMobile 2025.

Undergraduate Researcher

June 2023 – August 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Partnered to design and develop a program to operate multiple PX4 drones in a gazebo simulation through terminal controls, achieving successful control of up to 8 to 10 drones. (Python)
- Preprocessed two datasets, then trained and tested five ML models (KNN, SVM, Random Forest, Naive Bayes, Ensemble) for gesture recognition, enhancing classification accuracy by 10%. (Python)
- Co-authored a research paper for publication (DOI: [10.1145/3565287.3617621](https://doi.org/10.1145/3565287.3617621)) that documents findings and work, leading to publication in ACM and conference presentations. (LaTeX)

Projects

Gym Calendar Application (Memoranda) Modernization

- Designed and implemented a proof-of-concept integration with the Google Calendar API, enabling push of gym class schedules and resolving timezone alignment issues.
- Modernized the itinerary UI to improve usability, tailoring agenda views for students and trainers to display only relevant classes and sessions.
- Refactored portions of a 25,000+ LOC legacy Java codebase (Memoranda) to reduce build warnings and improve maintainability, while managing pull requests and resolving merge conflicts to streamline team collaboration.

Clustering Analysis of Steam Games Using Reviews, Sentiments, and Metadata

- Conducted clustering analysis on 42,497 Steam game records using K-Means and Agglomerative Clustering, implementing custom distance metrics (binary for categorical, Euclidean for numerical) to optimize performance, and documented the process and findings in an academic paper. (Python)
- Scraped and analyzed the sentiment of 385 games' English reviews using VADER and vectorized text with BERT embeddings, transforming Steamworks API JSON data into structured formats for clustering and analysis.
- Engineered a modular and reproducible Python workflow for end-to-end data collection, preprocessing, and analysis using pandas, scikit-learn, and transformers, with version control via Git and Git LFS for scalability and efficiency.

Console-Based Chat Application

- Designed and Organized local host pSQL database to store information about the user and messages, ensuring efficient data storage and queries. (Java, PostgreSQL)
- Developed console-based directions and options for users to register, log in/out, update account info, create a chat room, and etc., streamlining user interactions.